

Propositional logic, Introduction :

(Valid logical arguments)

Q. What is Proposition?

⇒ A Proposition is a declarative statement that is either true (T) or false (F), but not both.

* True = T = 1 (সত্যকে বোঝায়)

* False = F = 0 (সত্যকে বোঝায়)

Example: (Statement দেখা যায় True or False কিনা)
যদি True or False এ প্রকাশ করা যায়
তাহলে (এটি Proposition).

i. $1 + 1 = 2$

→ Truth value of this Proposition: T

ii. $2 + 2 = 5$

→ Truth value of this Proposition: F

iii. Sylhet is the capital of Bangladesh.

→ Truth value of this Proposition: F

iv. Pigs can fly.

→ Truth value of this Proposition: F

Not Proposition (not valid logical arguments):

Example:

i) What time is it? [Note: Questions বাক্যনা Proposition হতে পারে না,

ii) Read this carefully. [Note: Commands/Imperative

iii) Do your homework. বাক্যনা Proposition হতে পারে না]

iv) $x + 1 = 2$. [Note: Non constant values/value is not defined বাক্যনা Proposition হতে পারে না.]

v) Bangladesh and India. [Note: Not statements বাক্যনা Proposition হতে পারে না.]

vi) Rafi is the best lecturer. [Note: Opinion বাক্যনা Proposition হতে পারে না.]

vii) He is a college student. [Note: Person is not defined বাক্যনা Proposition হতে পারে না.]

Propositional Variables / statement variable :-

⊗ P/q/r/s
Propositional variable

① Today is Friday $\xrightarrow{\text{variable or statement}}$ $P = \text{Today is Friday}$

② It is raining $\xrightarrow{\text{" "}}$ $Q = \text{It is raining}$

The truth value of the propositional variable can be True or False.

Compound Proposition :-

⊗ Compound proposition is a proposition formed by combining two or more simple proposition.

⊗ The logical operators that are used to form compound propositions called connectives.

Symbol	Math name	English name
$\neg$	Negation	Not
$\vee$	Disjunction	OR
$\wedge$	Conjunction	AND
$\oplus$	EXOR	"OR... but not both"
$\rightarrow$	Implication / conditional	"If... then"
$\leftrightarrow$	Equivalence / biconditional	"If and only if"

← connective

Logical operator/connectives:

① **NOT** →
(negation)

Example:

Q. Find the negation of each of these Propositions

1. Today is Friday.

$\Rightarrow P = \text{Today is Friday}$

$\neg P = \text{It is not the case that today is Friday.}$

Truth table এর জন্য

* * * * * প্রতিটি চলক = 2^n ; n = চলক Variable

Truth table:

P	$\neg P$
F	T
T	F

2. 6 is negative.

$P = 6 \text{ is negative}$

$\neg P = \text{It is not the case that 6 is negative}$

3. $2+1=3$

$\neg P \rightarrow 2+1 \neq 3$

✓

∧

⊕

→

↔

4. There is no pollution in new jersey.

2. $\neg P$ = There is pollution in new jersey

5. The summer in marine is hot and sunny.

$\neg P$ = It is not the case that the summer in
marina is hot and sunny.

6. Today is Thursday.

TP = Today is not Thursday. / It is not the case that today is Thursday.

ii) AND $\longrightarrow$
(conjunction)

Example:

1. Today is Friday and it is raining.

$P = \text{Today is Friday}$

$q =$ It is raining

Truth table:

⊗ AND এ হাথোনা Input
যখন ০ True হলে তখনই
Output True হলে অন্যথায়
False হবে।

P	Q	$P \wedge Q$
F	F	F
F	T	F
T	F	F
T	T	T

note:

৭ (এখানে ৭ এ ১ এবং
 ৭ এ ২ জান মান ধরতে
 হবে, তাই যদি কোনো
 Variable থাকে তাহলে
 ২ এর double ৪ তাই
 variable থাকলে ৭ এর আরও
 ৪ এর সমান হবে দুইগুন হবে)

iii) OR $\longrightarrow$
(Disjunction)

Example:

[Con Example]

1. Today is Friday OR It is raining
P V q

P = Today is Friday

q = It is raining

logical expression $\longrightarrow$ $P \vee q$

Truth table:

* OR এর ফলাফল P ও q variable এর Input এর False হলে তখনই Output False হবে।

P	q	$P \vee q$
F	F	F
F	T	T
T	F	T
T	T	T

④ EXOR / Exclusive OR / XOR →

Example:

1. students who have taken calculus or computer science can take this class.

→ OR

Truth table:

[এটা OR এর Example;
গাণিতিক বা অন্য কৃত্রিম]

P	Q	$P \vee Q$
F	F	F
F	T	T
T	F	T
T	T	T

2. students who have taken calculus or computer science but not both can take the class.

→ XOR

Truth table:

⊕ বিচার্য অর্থাতঃ Input True হলে Output True.

P	Q	$P \oplus Q$
F	F	F
F	T	T
T	F	T
T	T	F

Note:
(যদি মোটামুটি
একটি সত্য হয়
হয়, অন্যটি মিথ্যা
হয় না হলে
EXOR)

Q. For each of these sentences, determine whether an exclusive OR or OR is intended.

a) coffee OR tea comes with dinner.

$\Rightarrow$ EXOR $\rightarrow P \oplus Q$

b) Experience with C++ OR Java is required

$\Rightarrow$ OR $\rightarrow P \vee Q$

c) lunch includes soup OR salad

$\Rightarrow$ EXOR $\rightarrow P \oplus Q$

d) you can pay using dollars OR euros.

$\Rightarrow$ OR / EXOR $\rightarrow \cancel{P \vee Q} / P \oplus Q$

① Implication/Conditional $\rightarrow$ "if p , then q "

Example:

1. p : Today is holiday.

q : The store is closed.

logical expression ~~and~~ $p \rightarrow q$

premise $\swarrow$ $\searrow$ conclusion

Truth table:

p	q	$p \rightarrow q$
F	F	T
F	T	T
T	F	F
T	T	T

Q. Determine whether each of these conditional statements is true or false.

a) If $1+1=2$, then $2+2=5$

$$\textcircled{\otimes} T \rightarrow F = F$$

$$\Rightarrow p \rightarrow q \quad / \quad T \rightarrow F = F$$

b) If $1+1=3$, then $2+2=4$ ① If $1+1=2$ then dogs can fly.

$$\Rightarrow F \rightarrow T = T$$

$$\Rightarrow T \rightarrow F = F$$

c) If $1+1=3$, then dogs can fly.

$$\Rightarrow F \rightarrow F = T$$



Other cases

~~⊗⊗~~ "If p, q "

= If It is below freezing, It is also snowing.
 $\Rightarrow p \rightarrow q$

⊗⊗ " p is sufficient for q "

= Driving over 65 miles per hour is ~~at~~ sufficient
for getting a speeding ticket.

$\Rightarrow p \rightarrow q$

⊗⊗ " q if p "

= Maria ~~is~~ will get a good job if she learns
Discrete Mathematics

$\Rightarrow$ ~~$p \rightarrow q$~~ $p \rightarrow q$

⊗⊗ " p implies q " $\therefore p \rightarrow q$

⊗⊗ " q whenever p " $\therefore p \rightarrow q$

⊗⊗ " q is necessary for p " $\therefore p \rightarrow q$

⊗⊗ " q follows from p " $\therefore p \rightarrow q$

⊗⊗ "a sufficient condition for q is p " $\therefore p \rightarrow q$

⊗⊗ " q when p " $\therefore p \rightarrow q$

~~⊗⊗~~ " q "

q whenever p $\therefore p \rightarrow q$
निश्चय मात्र।

⊗⊗ "q unless $\neg p$ "

= Maria will find a good job unless she does not learn Discrete Mathematics.

$$\Rightarrow p \rightarrow q$$

⊗⊗ "p only if q" :- $p \rightarrow q$

⊗⊗ "a necessary condition for p is q"

Ⓥ BICONDITIONAL / bi-implication :

Example:

1. You can take the flight if and only if you buy a ticket.

$$\Rightarrow p \leftrightarrow q$$

⊗⊗ "p is necessary and sufficient for q" :- $p \leftrightarrow q$

⊗⊗ "p iff q"

⊗⊗ "If p then q, and conversely :- $p \leftrightarrow q$ "

Truth table:

p	q	$p \leftrightarrow q$
F	F	T
F	T	F
T	F	F
T	T	T

⊗ यदि इन Input True/false के Output True रहे,

Q. Determine whether these biconditionals are True or False

a) $2+2=4$ if and only if $1+1=2$

$$= T \leftrightarrow T = T$$

b) $1+1=2$ if and only if $2+3=4$

$$= T \leftrightarrow F = F$$

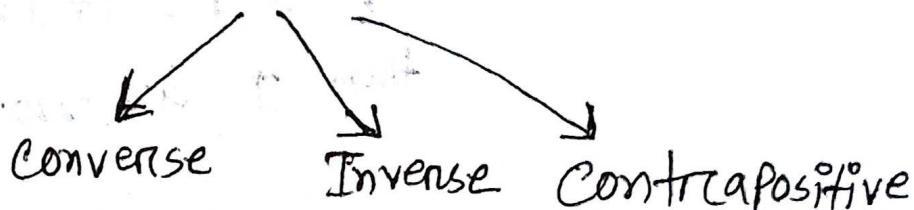
c) $1+1=3$ if and only if monkeys can fly.

$$= F \leftrightarrow F = T$$

$P \leftrightarrow Q$	P	Q
T	T	T
F	T	F
F	F	T
T	F	F

converse, Inverse, Contrapositive

Conditional $\rightarrow$ new con. statement.



(i) converse:

$$P \rightarrow Q = Q \rightarrow P \text{ (converse)}$$

(ii) Inverse:

$$P \rightarrow Q = \neg P \rightarrow \neg Q \text{ (Inverse)}$$

(iii) contrapositive:

contrapositive



converse + inverse

$$P \rightarrow Q = \neg Q \rightarrow \neg P$$

Example:

1. P : I am in Sylhet Q : I am in Bangladesh

= If I am in Sylhet then I am in Bangladesh.

$$= P \rightarrow Q$$

converse: $Q \rightarrow P$ = If I am in Bangladesh then I am in Sylhet.

Inverse: $\neg p \rightarrow \neg q =$ If I am not in sylhet then I am not in Bangladesh.

Contrapositive: $\neg q \rightarrow \neg p =$ If I am not in Bangladesh then I am not in sylhet.

Truth table:

p	q	$q \rightarrow p$	$\neg p$	$\neg q$	$\neg q \rightarrow \neg p$
F	F	T	T	T	T
F	T	F	T	F	T
T	F	T	F	T	F
T	T	T	F	F	T

$\neg p \rightarrow \neg q$	$p \rightarrow q$
T	T
F	T
T	F
T	T

Q. State the converse, inverse and contrapositive of each of these following statements:-

Q) If it snows today, I will ski tomorrow

$$\Rightarrow p \rightarrow q$$

converse: $q \rightarrow p$ = I will ski tomorrow only if it snows today

Inverse: $\neg p \rightarrow \neg q$ = If it does not snow today, then I will not ski tomorrow

contrapositive: $\neg q \rightarrow \neg p$ = ~~I will not ski tomorrow if it does not snow today.~~

Truth table of compound proposition:-

Example:

⊛⊛ Precedence of logical operators:
(যে অপারেটরগুলো আগে বাড়া প্রদান করা হবে):

$\neg, \wedge, \vee, \rightarrow, \leftrightarrow$

Example:

$$1. (P \vee \neg q) \rightarrow (P \wedge q)$$

Truth table:

P	q	$\neg q$	$P \vee \neg q$	$P \wedge q$	$(P \vee \neg q) \rightarrow (P \wedge q)$
F	F	T	T	F	F
F	T	F	F	F	T
T	F	T	T	F	F
T	T	F	T	T	T

$$2. (P \leftrightarrow Q) \oplus (P \leftrightarrow \neg Q)$$

Truth table:

P	Q	$\neg Q$	$P \leftrightarrow Q$	$P \leftrightarrow \neg Q$	$(P \leftrightarrow Q) \oplus (P \leftrightarrow \neg Q)$
F	F	T	T	F	T
F	T	F	F	T	T
T	F	T	F	T	T
T	T	F	T	F	T

$$3. (\text{practise}) \Rightarrow (\neg P \leftrightarrow \neg Q) \leftrightarrow (Q \leftrightarrow R)$$

English sentence $\rightarrow$ Proposition:

Q. Let P and Q be the Propositions -

P : It is below freezing

Q : It is snowing

Write these propositions using P and Q and logical connectives:

a) It is below freezing and snowing
 $= P \wedge Q$

b) It is below freezing but not snowing
 $= P \wedge \neg Q$

c) It is either snowing or freezing but not both.
 $= P \oplus Q$

d) If it is below freezing, it is also snowing.
 $= P \rightarrow Q$

e) It is either below freezing or it is snowing, but it is not snowing if it below freezing
 $= (P \vee Q) \wedge (P \rightarrow \neg Q)$

f) That is below freezing necessary and sufficient
for it to be snow

$$= P \leftrightarrow Q$$

g) you can access the internet from campus only if you are a computer science major or you are not a freshman

$$= P \rightarrow Q \vee \neg R$$

Proposition $\rightarrow$ English sentence

⊗ Q. let P and Q be the propositions "swimming at the new Jersey is allowed." and "sharks have been spotted near the shore." ~~respectively.~~

Express each of these compound propositions as an English sentence.

a) $\neg P$ = swimming at the new Jersey is not allowed.

b) $P \vee Q$ = swimming at the new Jersey is allowed or sharks have been spotted near the shore.

c) $P \rightarrow Q$ = If swimming at the new Jersey is allowed then sharks have been spotted near the shore.

Propositional equivalence:

$$\neg(P \wedge Q)$$

F
F
F
T

$\equiv$

$$\neg P \vee \neg Q$$

F
F
F
T

→ equivalence

⊗ ⊗ ⊗

Compound proposition

↙
Tautology

↘
contradiction

↘
contingency

Tautology, contradiction, contingency:

- (i) $P \vee \neg P$ (ii) $Q \wedge \neg Q$ (iii) $P \rightarrow Q$

Truth table:

P	Q	$\neg P$	$\neg Q$	$P \vee \neg P$	$Q \wedge \neg Q$	$P \rightarrow Q$
F	F	T	T	T	F	T
F	T	T	F	T	F	T
T	F	F	T	T	F	F
T	T	F	F	T	F	T

→ Tautology
(always true)

→ contradiction
(always false)

→ contingency
(mix)

Q. Show that $(P \rightarrow Q) \wedge (Q \rightarrow R) \rightarrow (P \rightarrow R)$ is a tautology using truth table.

$\Rightarrow$

P	Q	R	$P \rightarrow Q$	$Q \rightarrow R$	$P \rightarrow R$	$(P \rightarrow Q) \wedge (Q \rightarrow R)$
F	F	F	T	T	T	T
F	F	T	T	T	T	T
F	T	F	F	F	T	F
F	T	T	T	T	T	T
T	F	F	F	T	F	F
T	F	T	F	T	T	F
T	T	F	T	F	F	F
T	T	T	T	T	T	T

$(P \rightarrow Q) \wedge (Q \rightarrow R) \rightarrow (P \rightarrow R)$
T
T
T
T
T
T
T
T
T

~~The compound proposition~~ $(P \rightarrow Q) \wedge (Q \rightarrow R) \rightarrow (P \rightarrow R)$ is a tautology.

Q. Determine whether $(\neg P \wedge (P \rightarrow Q)) \rightarrow P$ is a tautology.

$\Rightarrow$

P	Q	$\neg P$	$P \rightarrow Q$	$(\neg P \wedge (P \rightarrow Q))$	$(\neg P \wedge (P \rightarrow Q)) \rightarrow P$
F	F	T	T	T	F
F	T	T	T	T	F
T	F	F	F	F	T
T	T	F	T	F	T

$(\neg P \wedge (P \rightarrow Q)) \rightarrow P$ is not a tautology.

Logical Equivalence using truth table:-

Example:-

~~Q.~~ Q. show that $P \rightarrow Q$ and $\neg P \vee Q$ are logically equivalent.

$\Rightarrow$

P	Q	$\neg P$	$P \rightarrow Q$	$\neg P \vee Q$
F	F	T	T	T
F	T	T	T	T
T	F	F	F	F
T	T	F	T	T

Using truth table

$P \rightarrow Q$ and $\neg P \vee Q$ are logically equivalent.

logically equivalent.

P	Q	$\neg P$	$P \rightarrow R$	$Q \rightarrow R$	$(P \rightarrow R) \vee (Q \rightarrow R)$	$P \wedge Q$
F	F	F	T	T	T	F
F	F	T	T	T	T	F
F	T	F	T	F	T	F
F	T	T	T	T	T	F
T	F	F	F	T	T	F
T	F	T	T	T	T	F
T	T	F	F	F	F	F
T	T	T	T	T	T	T

[illegible]

Logic Laws:-

i) Double negation law: $\neg(\neg P) \equiv P$

ii) Idempotent law: $P \vee P \equiv P$
 $P \wedge P \equiv P$

iii) Identity law:

$$P \wedge T \equiv P$$

$$P \vee F \equiv P$$

$$\left[\begin{array}{l} \text{Example:} \\ (P \wedge T) \wedge (P \vee F) \\ = P \wedge P \\ = P \end{array} \right]$$

iv) Domination law:

$$P \vee T \equiv T$$

$$P \wedge F \equiv F$$

$$\left[\begin{array}{l} (P \vee T) \wedge (P \wedge F) \\ = T \wedge F \\ = F \end{array} \right]$$

v) Commutative law:

$$P \vee Q \equiv Q \vee P$$

$$P \wedge Q \equiv Q \wedge P$$

vi) Associative law: $(P \vee Q) \vee R \equiv P \vee (Q \vee R)$
 $(P \wedge Q) \wedge R \equiv P \wedge (Q \wedge R)$

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Inverse law:-

$$P \vee \neg P \equiv T$$

$$P \wedge \neg P \equiv F$$

Inverse law:-

$$P \vee \neg P \equiv T$$

$$P \wedge \neg P \equiv F$$

$$A \bar{B} B \bar{C} + B \bar{C} C \bar{D} + \bar{A} D A \bar{B} + \bar{A} C D \bar{D}$$

$$= A \cdot 0 \cdot \bar{C} + B \cdot 0 \cdot \bar{D} + A \cdot 0 \cdot \bar{B} \cdot D + \bar{A} C \cdot 0$$

$$= 0$$

Absorption law:-



$$P \vee (P \wedge Q) \equiv P$$

$$P \wedge (P \vee Q) \equiv P$$

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Distributive law:-

$$PV(q \wedge r) \equiv (PVq) \wedge (PVr)$$

$$P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$$



conditional law:-

$$P \rightarrow Q \equiv \neg P \vee Q$$

Biconditional law:-

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

★★★★★

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De Morgan's law:-

$$\neg(P \wedge Q) \equiv \neg P \vee \neg Q$$

$$\neg(P \vee Q) \equiv \neg P \wedge \neg Q$$

$$\textcircled{2} \quad \neg(R \wedge S \wedge T \wedge Q) \equiv \neg R \vee \neg S \vee \neg T \vee \neg Q$$

$$(*) \quad \neg(R \vee S \vee Q) \equiv \neg R \wedge \neg S \wedge \neg Q$$

$$\therefore \neg(R \wedge S \wedge T) \wedge \neg(R \vee S \vee T)$$

$$= (\neg R \vee \neg S \vee \neg T) \wedge (\neg R \wedge \neg S \wedge \neg T)$$

$$= (\bar{R} + \bar{S} + \bar{T}) \wedge (\bar{R} \cdot \bar{S} \cdot \bar{T})$$

$$= \bar{R} \bar{R} \bar{S} \bar{T} + \bar{R} \bar{S} \bar{T} \bar{S} + \bar{R} \bar{S} \bar{T} \bar{T}$$

$$= \bar{R} \bar{S} \bar{T} + \bar{R} \bar{S} \bar{T} + \bar{R} \bar{S} \bar{T}$$

$$= \bar{R} \bar{S} \bar{T}$$

$$= \neg R \wedge \neg S \wedge \neg T$$

De Morgan's law to find the negations

Q. Use De Morgan's law to find the negation of each of the following statements.

a) Jon is rich and happy

$$\Rightarrow \neg(P \wedge Q) = \neg P \vee \neg Q = \text{Jon is not rich or not happy.}$$

b) Carlos will bicycle or run tomorrow.

$$\Rightarrow \neg(P \vee Q) = \neg P \wedge \neg Q = \text{Carlos will not bicycle and not run tomorrow.}$$

~~Q.~~ Q. Show that each of these conditional statements is a tautology without using truth table.

$$\textcircled{a} (P \wedge Q) \rightarrow P$$

$$\equiv \neg(P \wedge Q) \vee P$$

$$\equiv \neg P \vee \neg Q \vee P$$

$$\equiv \neg P \vee P \vee \neg Q$$

$$\equiv T \vee \neg Q$$

$$\equiv T \rightarrow (\text{Tautology})$$

$$\textcircled{b} (P \wedge Q) \rightarrow (P \vee Q)$$

$$\equiv \neg(P \wedge Q) \vee (P \vee Q)$$

$$\equiv \neg P \vee \neg Q \vee P \vee Q$$

$$\equiv (\neg P \vee P) \vee (\neg Q \vee Q)$$

$$\equiv T \vee T$$

$$\equiv T \rightarrow (\text{Tautology})$$

Q. Show that each of statement are logically equivalent without using truth table.

$$a) \neg(P \rightarrow Q) \equiv P \wedge \neg Q$$

$$L.H.S = \neg(P \rightarrow Q)$$

$$= \neg(\neg P \vee Q)$$

$$= \neg(\neg P) \wedge \neg Q$$

$$= P \wedge \neg Q$$

$$= R.H.S$$

~~$$\textcircled{a} \neg P(P \rightarrow Q)$$~~

$$\textcircled{b} \neg(P \vee (\neg P \wedge Q)) \equiv \neg P \wedge \neg Q$$

$$L.H.S = \neg(P \vee (\neg P \wedge Q))$$

$$\equiv \neg P \wedge \neg(\neg P \wedge Q)$$

$$\equiv \neg P \wedge [\neg(\neg P) \vee \neg Q]$$

$$\equiv \neg P \wedge (P \vee \neg Q)$$

$$\equiv (\neg P \wedge P) \vee (\neg P \wedge \neg Q)$$

$\equiv F \vee (\neg P \wedge \neg Q)$

$\equiv \neg P \wedge \neg Q$